

IKT215 Pattern Recognition: Final Project

Posted on: April 24th, 2026 00:01.

Deadline: May 18th, 2026 14:00.

Instructions

1. Read these instructions and follow them for your submission.
2. Prepare and submit a report (maximum 15 pages, PDF format) containing:
 - Introduction and Background,
 - Methodology,
 - Results and Discussions,
 - Conclusion.
3. Report quality and structure will be awarded 10% of the total marks.
4. You don't need to submit your code.
5. Report and justify all hyperparameter choices you make.
6. All visualizations must be legible: use readable font sizes, labelled axes etc. Screenshots of console output are not acceptable as figures.
7. **Your project report should be uploaded in PDF format to the designated CANVAS folder before the deadline.**

Use of Generative AI:

Use of generative AI tools is strongly discouraged for this project. All reports **must include a dedicated disclaimer page** regarding the use of generative AI tools in accordance with [UiA's general guidelines](#). **Submitting AI-generated content without proper citation constitutes plagiarism and will be treated as such.**

Dataset Description

The Fashion MNIST dataset provides 70,000 grayscale clothing images (28×28 pixels) across 10 categories. It features simplified black-and-white sketches of garments and accessories like T-shirts, trousers, dresses, ankle boots etc.

Import Dataset

1. PyTorch

```
from torchvision import datasets
from torchvision.transforms import ToTensor
```

```
train_data = datasets.FashionMNIST(
    root="data",
    train=True,
    download=True,
    transform=ToTensor()
)
```

```
test_data = datasets.FashionMNIST(
    root="data",
    train=False,
    download=True,
    transform=ToTensor()
)
```

2. Tensorflow

```
import tensorflow as tf

(train_images, train_labels), (test_images, test_labels) =
tf.keras.datasets.fashion_mnist.load_data()
```

Objective

Evaluate dense neural networks against k -Nearest Neighbours (k NN) algorithm on Fashion MNIST dataset. **Use the held-out test set only for final reporting.**

Tasks (90% Marks)

Task 1: Exploratory Analysis (15% Marks)

- Import and analyze the Fashion MNIST dataset: 60,000 images for training/validation, and 10,000 images for testing.
- Include in your report:
 - Describe any preprocessing steps you applied to the data before modelling. Justify your choices. (5%)
 - Visualization of one random sample each from different classes with corresponding labels. (5%)
 - Visualization of class frequencies using bar charts and report the class distribution in the dataset. (5%)

Task 2: Benchmark Classifier (20% Marks)

- Implement a benchmark classifier using k NN algorithm to get a benchmark performance.
- Use 5-fold cross-validation on the training set to select the optimal value of k , distance metric, and weighting scheme. E.g. $k \in \{1, 3, 5, 7, 9, 15, 21\}$, distance metric $\in \{\text{Euclidean}, \text{Manhattan}\}$, weighting $\in \{\text{Uniform}, \text{Distance}\}$
- Include in your report:
 - Your choice of k -value, distance metric, and weighting. Justify your choices using cross-validation results. (6%)
 - The performance of your benchmark model on the test dataset and discuss: (6%)
 - Accuracy,
 - Precision, and
 - Recall.
 - A clearly labelled heatmap of the confusion matrix on the test set. (4%)
 - A discussion on the most misclassified classes. Propose a reason. (4%)

Task 3: Principal Component Analysis (15% Marks)

- Apply PCA to reduce the original 784 features to 5, 50, and 150 components.
- Retrain the k NN classifier at each dimensionality using the exact k , distance metric, and weighting found in Task 2 without retuning.
- Also evaluate k NN on the full 784-dimensional data (no PCA) as a baseline: this gives you four configurations in total: 784 (baseline), 150, 50, and 5 components.
- Include in your report:
 - Plot of classification accuracy vs. number of principal components/features. Discuss your key takeaways from this plot. (5%)

- Plot of total runtime (PCA + training) vs. accuracy for all four configurations. Discuss your key takeaways from this plot. (5%)
- Discuss tradeoffs between total runtime and model accuracy in selecting the principal components to work with. (5%)

Task 4: Neural Network (30% Marks)

- Implement a dense neural network (with ≥ 2 hidden layers) using the original 784-dimensional data.
- Implement validation error minimization model selection strategy.
- Consider at least 2 model architectures such as [256, 128], [512, 256], etc.
- In your report:
 - Discuss the model selection procedure: model architectures and hyperparameters considered. (6%)
 - Present the hyperparameters used in training the final model as a table and discuss their significance: (6%)
 - Architecture of the NN,
 - Learning rate,
 - Activation function,
 - Loss criterion, and
 - Optimizer type.
 - A plot of the training-validation loss curves. Discuss your key takeaways from this plot. (6%)
 - The performance of the trained model on the test dataset and discuss why the performance is acceptable: (6%)
 - Accuracy,
 - Precision, and
 - Recall.
 - A clearly labelled heatmap of the confusion matrix on the test set. Discuss which classes your NN model struggles with the most and why. (6%)

Task 5: Comparative Analysis (10% Marks)

- Compare and report the neural network performance with your benchmark model (784-d). Discuss which one outperforms the other and propose a reason. (5%)
- Discuss at least one concrete way in which the classifier can be improved further. (5%)

Report Quality and Structure (10% Marks)

- Introduction and Background: clear problem statement, objective. (1%)
- Methodology: description of methods, hyperparameter choices. (3%)
- Results and Discussions: analysis, visualization, and critical evaluation (4%)
- Conclusion: Summary of your work. (2%)